

The overall quality of disclosed θ is lower and buyers may be misled by fake signals. When $p < 1/2$, even though every seller is on the market (which is pareto efficient), the DMV is cheating and giving a fake signal more than half the time.

In practice, it is unrealistic for the regulator to provide signals with such poor quality. However, in this model note that any nonzero value of $p < 1$ achieves strictly greater profit for the DMV than the profit from the baseline perfect signal. Even an almost perfect signal $p = 1 - \epsilon$ achieves higher profit than $p = 1$. Thus, the DMV can exploit the information asymmetry in the market and increase its profit with imperfect signals.

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References

- [1] Akerlof George (1970) The market for ‘Lemons’: quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84:488–500